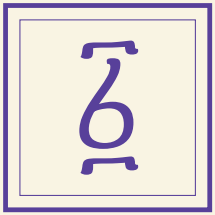
Business cases

The inner product and nearest neighbors

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026



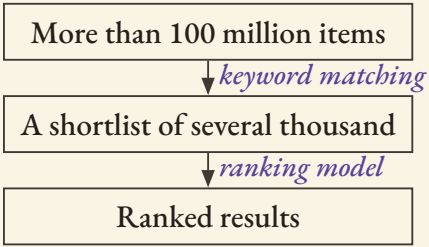
Section 1

Etsy

Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.

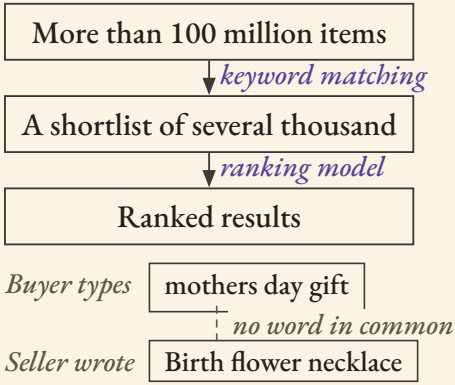


Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

Etsy: a shortlist out of 100 million items

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Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

The training data: a month of search logs

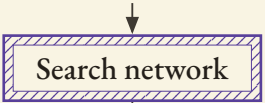
1 month: more than 150 million rows, from up to 30 million users.

Search	Who searched	Item	Action
mothers day gift	Zurich, searched necklaces	Flower necklace	purchase
bridge photos	New York, clicked print shops	Bridge print	cart add
wool socks	Oslo, searched knitting	Knitted socks	cart add

The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.

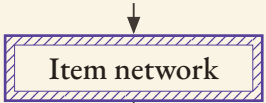
Every search and every item becomes a list of numbers

The search words and who searched



mothers day gift, in Zurich

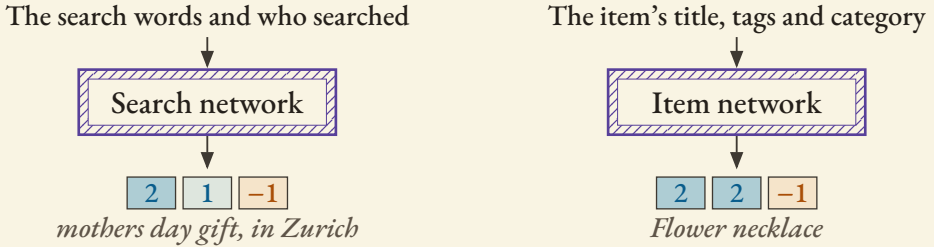
The item's title, tags and category



Flower necklace

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

Every search and every item becomes a list of numbers



Training makes the inner product high for items bought or added to the cart after the search: here 7.

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

A search arrives: sort by inner product

mothers day gift 2 1 −1

Invented numbers.

A search arrives: sort by inner product

mothers day gift	2	1	−1	*inner product*
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

Sorted: the top two join the shortlist

Flower necklace	7
Recipe board	4
Photo mug	3
Gift wrap	−4

Cost and value

Cost

72 hours *on 4 GPUs*

1 month *of search logs*

Value: A/B tests, relative change

+2.63% *visits that end in a purchase*

+5.58% *searches that end in a purchase*

Jha and others (Etsy), IEEE FMLDS 2024. Airbnb: arXiv preprint 2026.

Cost and value

Cost

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Airbnb calls its +0.31% gain “a large portion of the team’s annual goal”.

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Cost and value

Cost

72 hours *on 4 GPUs*

1 month *of search logs*

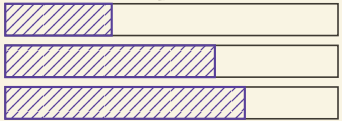
Value: A/B tests, relative change

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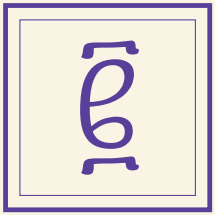
+5.58% *searches that end in a purchase*

Airbnb calls its +0.31% gain “a large portion of the team’s annual goal”.

Popular searches: share of the bought products found in the top 100



Jha and others (Etsy), IEEE FMLDS 2024. Airbnb: arXiv preprint 2026.

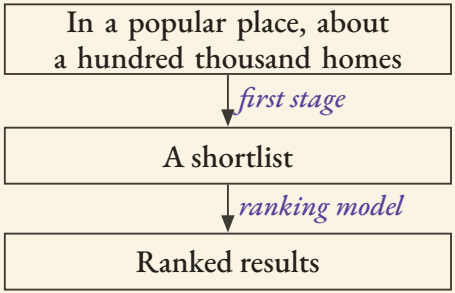


Section 2

Airbnb

Airbnb: the first stage of search

The first stage evaluates tens of thousands of homes.

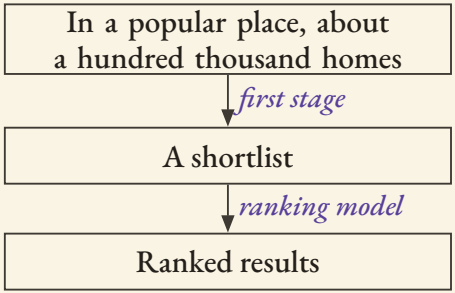


Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Airbnb: the first stage of search

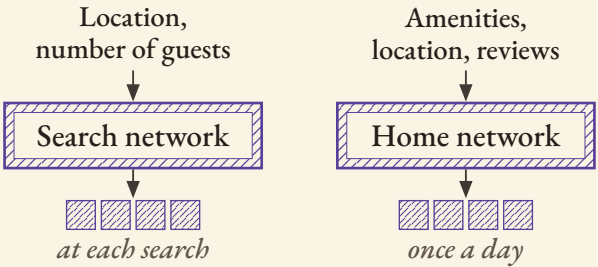
The first stage evaluates tens of thousands of homes.

It used to be only a simple linear model that scored every home once a day, whatever the search.



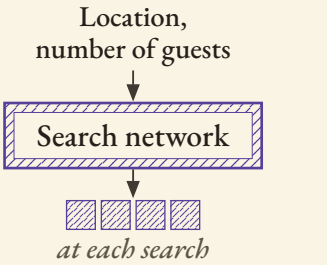
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes

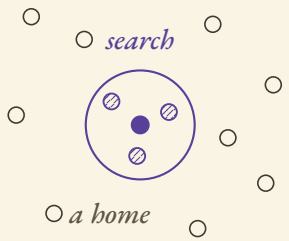


Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes

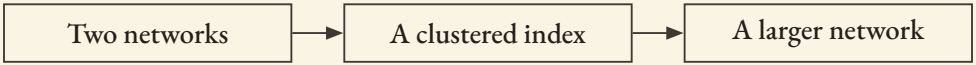


The nearest homes join the shortlist



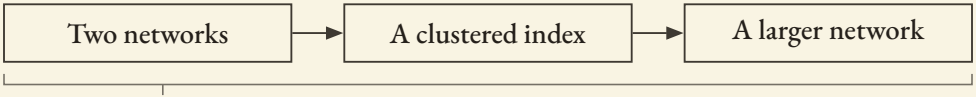
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



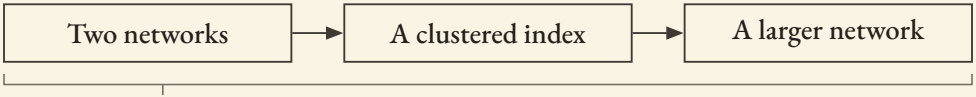
+0.31%

visits that end in a booking, all three, relative change

Airbnb: “a large portion of the team’s annual goal”.

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.

The value: three A/B tests



+0.31%

visits that end in a booking, all three, relative change

Airbnb: “a large portion of the team’s annual goal”.

+2.3% *bookings from promotional emails*

−16% *compute with the clustered index*

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.